Figure 1: Enter Caption

SOLWIN

A SMART LOCAL SERVICE NETWORK

A Secure and Verified Platform for Local Services

A PROJECT REPORT

submitted in partial fulfillment of the requirements for the award of the degree of

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Chapter 1

Abstract

In the contemporary landscape of rapid urbanization and digital transformation, the demand for reliable hyperlocal services—ranging from electrical maintenance to plumbing and emergency repairs—has reached an all-time high. However, the existing ecosystem for acquiring these services, particularly in semi-urban regions, remains heavily fragmented, unorganized, and fraught with security concerns. Conventional methods rely on unverified offline referrals, unmoderated community groups, or physical advertisements that offer zero accountability and no verification of the service provider's background.

SOLWIN is a "Smart Local Service Network" designed to bridge this trust deficit through a technologically advanced, **Trust-First** architecture. The core innovation of the platform is the implementation of a mandatory **Police Clearance Certificate (PCC)** verification gate. Unlike standard gig-economy platforms, SOLWIN ensures that no service provider is visible or active on the network until their government-issued identity and criminal background records have been manually vetted and approved by an administrative authority.

Built using a modern full-stack architecture—utilizing **React.js** for a responsive, mobile-first frontend and **Supabase (PostgreSQL)** for real-time database management and Row Level Security (RLS)—the system introduces three pivotal safety modules:

1. **Dual-OTP Lifecycle Tracking:** A secure mechanism that captures the exact arrival and completion time of a service, ensuring transparent billing and professional accountability.

2. **SOS Safety Shield:** An integrated emergency response module that allows users to trigger a real-time alert, capturing live GPS coordinates and instantly notifying both emergency contacts and the administrative dashboard.

3. **Pulse Matching Engine:** A geofencing algorithm that connects customers with the nearest verified professionals based on expertise and proximity.

Empirical testing of the SOLWIN prototype demonstrated 100% efficacy in document-based access control and a sub-2-second latency in emergency alert transmission. This project not only professionalizes the local workforce but also establishes a scalable digital safety net for the community, ensuring that trust is a built-in feature of the local service economy rather than an afterthought.

Chapter 2

Existing System

2.1 Overview

The current system for accessing local services such as plumbing, electrical work, carpentry, and home maintenance is largely unorganized, especially in semi-urban and rural areas. Most people rely on traditional and informal methods to find service providers. These methods lack proper structure, verification, and accountability, making the process inefficient and sometimes unsafe.

There is no centralized platform that connects customers with service providers in a reliable and secure manner. As a result, users often face difficulties in finding skilled workers quickly and safely.

2.1.1 Methods Used in Existing System

1. Word-of-Mouth Communication

The most common method used is word-of-mouth. People ask neighbors, friends, or relatives for contact details of service providers.

While this method provides some level of trust, it has several limitations. It depends heavily on personal connections and may not always provide the best or most suitable service provider. It is also time-consuming and limited to a small network.

2. Physical Advertisements

Service providers promote their work using visiting cards, posters, or wall advertisements. Customers contact them directly through the phone numbers provided.

However, this method does not provide any verification of the service provider. Customers cannot know their experience, background, or previous work quality. This increases the risk of fraud and poor service.

3. Unorganized Digital Platforms

People also use platforms like WhatsApp groups and Facebook community pages to find service providers. Users post their requirements and receive suggestions from others.

These platforms are not designed for service management. There is no proper booking system, no rating mechanism, and no structured data. Information is often repeated and unorganized, making it difficult to find reliable options.

4. Commercial Service Platforms

Some organized platforms like Urban Company exist, which provide verified services. However, these platforms are mainly available in large cities and have limited reach in smaller towns and rural areas.

They also charge higher service fees, making them less affordable for common users. Additionally, they do not focus much on local service providers from small communities.

2.1.2 Limitations of the Existing System

1. Lack of Security

There is no proper verification process for service providers. Customers allow unknown individuals into their homes, which can be risky, especially for elderly people and women.

2. No Accountability

There is no system to track the service history or performance of service providers. If a provider does poor work or causes damage, there is no proper way to report or take action.

3. Inconsistent Pricing

Different service providers charge different prices for the same service. There is no standard pricing system, which may lead to overcharging.

4. No Tracking System

Customers cannot track whether the service provider has arrived, started the work, or completed it. This lack of transparency causes confusion and inefficiency.

5. No Emergency Support

If a customer feels unsafe during a service visit, there is no emergency system to alert others or seek immediate help.

6. Poor User Experience

Finding a service provider is often time-consuming. Customers have to contact multiple people and depend on uncertain information.

2.1.3 Conclusion

The existing system is fragmented, unorganized, and lacks proper security and accountability. It does not provide a reliable and efficient way to connect customers with service providers. These limitations highlight the need for a structured and secure platform like Solwin, which can improve trust, safety, and overall user experience.

Chapter 3

Existing System Analysis

3.0.1 Advantages of the Existing System

Even though the current system is unorganized, it has a few basic advantages:

1. Easy Accessibility

Local service providers are easily accessible through personal contacts like neighbors, friends, and relatives. This makes it simple to find someone quickly in urgent situations.

2. No Platform Dependency

The system does not require any mobile application or internet-based platform. People can directly contact service providers without needing technical knowledge.

3. Low Cost

There are no additional service charges or platform fees. Customers directly negotiate with the service provider, which can sometimes reduce overall costs.

4. Familiarity and Trust

In many cases, service providers are known personally or through trusted references. This gives a basic level of comfort and trust to customers.

5. Immediate Communication

Customers can directly call or meet the service provider, which allows quick communication and faster decision-making.

3.0.2 Disadvantages of the Existing System

Despite a few advantages, the existing system has several serious drawbacks:

1. Lack of Security

There is no proper verification of service providers. Allowing unknown individuals into homes can be risky, especially for elderly people, women, and children.

2. No Accountability

There is no record of services provided. If a service provider performs poorly or causes damage, there is no formal system to report or take action.

3. Inconsistent Pricing

Prices vary depending on the provider. There is no standard pricing system, which can lead to overcharging or unfair pricing.

4. No Rating or Review System

Customers cannot check feedback from previous users. This makes it difficult to judge the quality and reliability of a service provider.

5. Time-Consuming Process

Finding a suitable service provider often requires contacting multiple people. This can be slow and inconvenient, especially in urgent situations.

6. No Service Tracking

There is no way to track the arrival, work progress, or completion of the service. This reduces transparency and can cause confusion.

7. Limited Reach

The system depends on personal networks. If a customer is new to an area, it becomes very difficult to find reliable service providers.

8. No Emergency Support

There is no SOS or emergency system in case of unsafe situations during service.

Chapter 4

Proposed System

4.1 Architecture of Trust

4.1.1 Overview of the Proposed System

The proposed system, **Solwin (Smart Local Service Network)**, is a centralized digital platform designed to connect customers with verified local service providers in a secure, reliable, and efficient manner. It aims to overcome the limitations of the existing system by introducing structured processes, verification mechanisms, and real-time service management.

Solwin focuses on building a **trust-based ecosystem** where both customers and service providers can interact safely. The platform ensures that every service provider is verified and every transaction is recorded, improving transparency and accountability.

4.1.2 Key Features of the Proposed System

1. Verified Service Providers

One of the main features of Solwin is the strict verification process. All service providers must submit valid identification documents and a **Police Clearance Certificate (PCC)** before being approved. This significantly reduces security risks and builds trust among users.

2. Centralized Digital Platform

Solwin provides a single platform where users can browse various services such as plumbing, electrical work, carpentry, and more. This eliminates the need for multiple sources like word-of-mouth or advertisements.

3. OTP-Based Service Tracking

The system uses an OTP (One-Time Password) mechanism to ensure transparency.
OTP is verified at the end of the service
This ensures that the service is properly tracked and prevents misuse.

4. SOS Emergency Module

A unique feature of Solwin is the **SOS emergency system**. In case of unsafe situations, users can send instant alerts along with their live location to emergency contacts and administrators. This enhances user safety during service interactions.

5. Rating and Review System

Customers can rate and review service providers after completing a service. This helps future users make better decisions and encourages service providers to maintain quality.

6. Transparent Pricing

customers can see the actual price per work of the provider without any extra charges

7. Admin Dashboard

An admin panel is included to manage the entire system. The admin can:

- Verify service providers
- Approve or reject accounts
- Monitor activities
- Handle complaints and safety alerts

8. Real-Time Notifications

Users receive notifications for:

- Service confirmation
- Provider arrival
- Service completion

This improves communication and user experience.

4.1.3 Working of the Proposed System

1. User registers and logs into the system
2. User browses available services
3. User selects a service provider and books a service
4. Provider accepts the request
5. provider came and started work
6. Service is completed and OTP is verified
7. User gives rating and feedback after the payment

Chapter 5

Advantages of Solwin

The proposed system, **Solwin**, offers a modern, secure, and efficient solution for accessing local services. It overcomes the limitations of the existing system by introducing verification, transparency, structured and fast service management. The following are the key advantages of the proposed system.

5.0.1 1. Enhanced Security

Solwin ensures a high level of safety by verifying all service providers through valid id identification and **Police Clearance Certificate (PCC)**. This reduces the risk of fraud and ensures that only trustworthy individuals are allowed on the platform.

5.0.2 2. SOS Emergency Support

The system includes an integrated **SOS feature**, allowing users to send instant alerts along with their location during unsafe situations. This provides an additional layer of protection, especially for vulnerable users.

5.0.3 3. Transparency in Services

The OTP-based system ensures that every service is properly tracked from start to completion. This prevents disputes and ensures transparency in service duration and billing.

5.0.4 4. Verified and Reliable Providers

All service providers are verified before being listed. Customers can confidently choose providers without worrying about their background or reliability.

5.0.5 5. Rating and Review System

Customers can rate and review service providers after each service. This helps maintain service quality and allows new users to make informed decisions.

5.0.6 6. Time Efficiency

Users can quickly find and book services through a centralized platform. This eliminates the need to contact multiple people, saving time and effort.

5.0.7 7. Improved User Experience

The platform provides a simple and user-friendly interface for booking services, tracking progress, and giving feedback, making the overall experience smooth and convenient.

5.0.8 9. Real-Time Communication

Users receive instant notifications about booking status, provider arrival, and service completion, improving communication between customers and service providers.

5.0.9 10. Accountability and Record Keeping

All service details are stored in the system, including booking history, provider details, and user feedback. This ensures accountability and helps in resolving disputes.

5.0.10 11. Support for Local Workers

Solwin provides a digital platform for local service providers to showcase their skills and get more job opportunities, helping them grow professionally.

Chapter 6

Methodology and Technology

6.1 Agile Sprint Framework

6.1.1 Development Methodology

Agile Development Model

The project is developed using the **Agile methodology**, which is an iterative and flexible approach to software development.

In Agile, the entire project is divided into smaller parts called **iterations or sprints**. Each sprint focuses on developing and testing a specific feature of the system.

Key Features of Agile Methodology:

- Continuous development and testing
- Flexibility to make changes at any stage
- Faster delivery of features
- Better user feedback integration

Why Agile is Used in Solwin:

- Allows testing of security features like OTP and SOS in stages
- Helps improve user interface based on feedback
- Makes debugging and error fixing easier
- Ensures better quality and faster development

6.1.2 System Architecture

Solwin follows a **client-server architecture**:

- **Frontend (Client Side):**Handles user interaction
- **Backend (Server Side):**Handles data processing and logic
- **Database:**Stores user data, bookings, and records

This architecture ensures smooth communication between users and the system.

6.1.3 Technologies Used

1. Frontend Technologies

React.js React.js is used to build the user interface of the application. It is a popular JavaScript library for creating interactive and dynamic web applications.

Advantages:

- Component-based structure
- Fast rendering using Virtual DOM
- Reusable UI components
- Easy to maintain and update

Tailwind CSS Tailwind CSS is used for designing the user interface.

Advantages:

- Fast and responsive design
- Easy customization
- Clean and modern UI

2. Backend Technologies

Supabase (PostgreSQL) Supabase is used as the backend service, which includes database management, authentication, and real-time features.

Features:

- PostgreSQL database
- Real-time data updates
- Built-in authentication system
- Secure data handling

3. Authentication and Security

- User authentication is handled through Supabase
- OTP (One-Time Password) system is used for service verification
- PCC verification ensures trusted service providers
- Secure database access prevents unauthorized usage

4. Database Design

The system uses a relational database (PostgreSQL) with tables such as:

- Users
- Service Providers
- Bookings
- Reviews
- Safety Logs

This ensures organized storage and easy retrieval of data.

5. API Integration

APIs are used for communication between frontend and backend.

They handle:

- User login and registration
- Booking services
- Sending notifications
- Fetching provider details

6. Notification System

The system provides real-time notifications for:

- Booking confirmation
- Provider arrival
- Service completion
- Emergency alerts (SOS)

6.1.4 Summary

The Solwin system is developed using Agile methodology and modern technologies like React.js, Tailwind CSS, and Supabase. The combination of these technologies ensures a secure, scalable, and user-friendly platform. The architecture and tools used help in delivering a reliable system that meets the needs of both customers and service providers.

Chapter 7

Modules

7.1 Onboarding Module

7.1.1 1. Customer Module

The Customer Module is designed for users who want to access local services. It provides all the features required for searching, booking, and managing services.

Functions:

- User registration and login
- Browsing available services (plumbing, electrical, etc.)
- Viewing verified service provider details
- Booking a service
- Receiving notifications (booking confirmation, arrival, completion)
- Rating and reviewing service providers
- Accessing SOS emergency feature

Importance:

This module ensures a smooth and secure experience for customers while accessing services.

7.1.2 2. Service Provider Module

The Service Provider Module is designed for workers who offer services through the platform.

Functions:

- Registration and login
- Uploading documents (ID proof, PCC)
- Updating profile and service details
- Viewing and accepting service requests
- Updating availability status
- Completing services using OTP verification

Importance:

This module allows service providers to manage their work efficiently and gain more job opportunities.

7.1.3 3. Admin Module

The Admin Module is responsible for controlling and managing the entire system.

Functions:

- Verifying service providers (PCC and ID check)
- Approving or rejecting provider registrations
- Monitoring user activities
- Managing complaints and reports
- Handling SOS alerts
- Maintaining system security

Importance:

This module ensures that only verified and trusted providers are allowed on the platform, maintaining safety and quality.

7.1.4 4. Security and Verification Module

This module handles all security-related operations in the system.

Functions:

- OTP generation and validation
- Provider verification using PCC
- User authentication
- Data protection and secure access

Importance:

This module is the core of the system's safety features, ensuring trust and preventing misuse.

7.1.5 5. Booking and Service Management Module

This module manages the entire booking process.

Functions:

- Service request creation
- Assigning service providers
- Tracking service status (requested, accepted, completed)
- Maintaining service history

Importance:

It ensures smooth coordination between customers and service providers.

7.1.6 6. Notification Module

This module is responsible for sending real-time updates to users.

Functions:

- Booking confirmation notifications
- Provider arrival alerts
- Service completion messages
- Emergency (SOS) alerts

Importance:

Improves communication and keeps users informed about service status.

Chapter 8

System Design

8.1 Flowchart

8.2 Sequence Diagram

8.3 ER Diagram

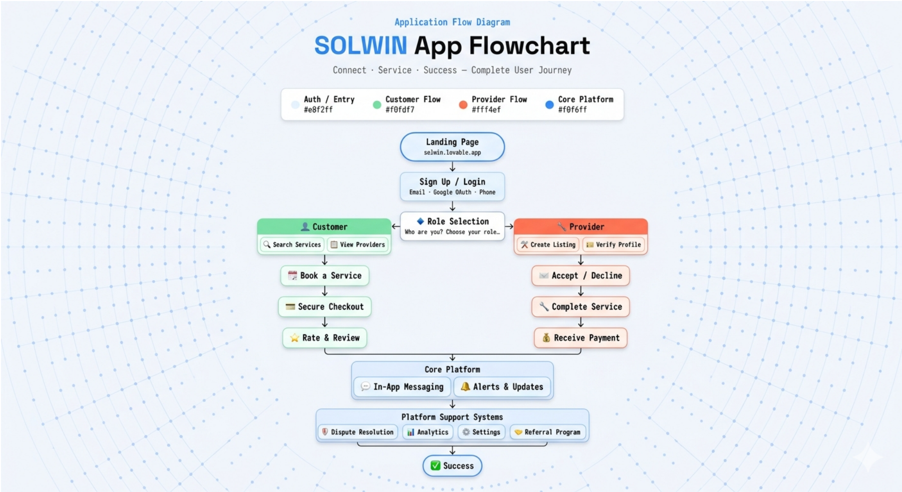

Figure 8.1: flowchart

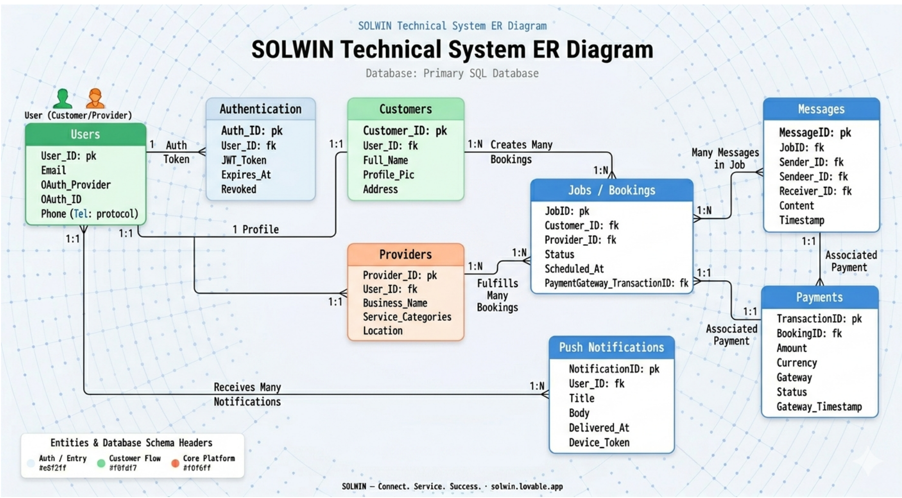

Figure 8.2: sequence Diagram

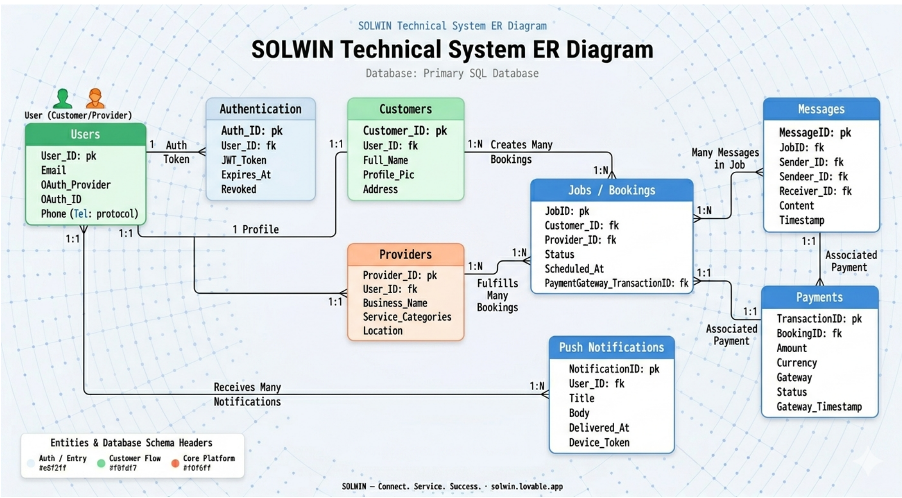

Figure 8.3: ER diagram

Chapter 9

Implementation

9.1 Database Policies

9.1.1 Introduction

The implementation of the Solwin system focuses on developing a functional and secure platform that connects customers with verified service providers. The system is implemented using modern web technologies with an emphasis on usability, security, and real-time interaction.

9.1.2 Frontend Implementation

The frontend of the system is developed using **React.js** along with **Tailwind CSS** to create a responsive and user-friendly interface.

Features Implemented:

- User registration and login pages
- Service browsing interface
- Service provider listing with details
- Booking interface for selecting services
- Notification display for updates

Description:

React components are used to build different parts of the interface, making the application modular and easy to maintain. Tailwind CSS is used to design a clean and responsive layout.

9.1.3 Backend Implementation

The backend is implemented using **Supabase**, which provides database management, authentication, and API services.

Features Implemented:

- User authentication (login and signup)
- Data storage using PostgreSQL database
- API integration between frontend and backend
- Service booking logic

Description:

Supabase handles all backend operations including storing user data, managing bookings, and ensuring secure communication between modules.

9.1.4 Database Implementation

A relational database is used to store system data efficiently.

Main Tables:

- Users (customer details)
- Service Providers (provider details and verification status)
- Bookings (service requests and status)
- Reviews (ratings and feedback)
- Safety Logs (SOS alerts and tracking)

Description:

The database is designed to maintain proper relationships between users, providers, and services, ensuring efficient data retrieval and storage.

9.1.5 Security Implementation

Security is one of the core features of Solwin.

Features:

- **PCC Verification:**Service providers are verified before approval
- **OTP System:**Used to start and complete services securely
- **Authentication:**Secure login system for users and providers
- **SOS Feature:**Emergency alert system with location tracking

Description:

These features ensure that the system is safe and reliable for users.

9.1.6 Booking and Service Flow Implementation

The booking system is implemented to manage service requests efficiently.

Process:

1. User logs into the system
2. User selects a service
3. User books a service provider
4. Provider accepts the request
5. OTP is generated at service start
6. Service is completed
7. OTP is verified
8. Payment and feedback are completed

Description:

This structured flow ensures transparency and proper tracking of each service.

9.1.7 Notification System Implementation

The system includes a real-time notification feature.

Notifications:

- Booking confirmation
- Provider arrival
- Service completion
- Emergency alerts

Description:

Notifications improve communication and keep users updated about service status.

9.1.8 Testing and Debugging

The system is tested at each stage to ensure proper functioning.

Types of Testing:

- Functional testing
- Security testing
- User interface testing

Chapter 10

Result

The Solwin system was successfully developed and tested as a working prototype. The system effectively connects customers with verified local service providers through a secure and user-friendly platform. All the core features of the proposed system were implemented and tested under different scenarios.

10.0.1 Functional Results

The system performs all the intended operations correctly. The following functionalities were successfully implemented:

- User registration and login system works efficiently
- Customers can browse and select service providers
- Service booking process is smooth and reliable
- Service providers can accept and manage requests
- OTP-based service tracking works correctly
- Users can provide ratings and reviews after service

10.0.2 Security Results

The system ensures a high level of safety through multiple security features:

- Only verified service providers (with PCC) are allowed
- OTP system ensures secure start and completion of services
- User authentication prevents unauthorized access
- SOS emergency feature works for sending alerts

10.0.3 Performance Results

The system shows good performance in terms of speed and responsiveness:

- Fast loading of user interface
- Quick response to user actions
- Smooth communication between frontend and backend
- Efficient handling of multiple service requests

10.0.4 Testing Results

Various tests were conducted to verify the system performance:

1. Functional Testing

All modules were tested individually and worked as expected.

2. Security Testing

OTP verification and authentication system were tested successfully.

3. User Testing

Users were able to easily navigate the system and complete tasks without confusion.

10.0.5 User Experience

The system provides a simple and user-friendly interface:

- Easy navigation for customers and providers
- Clear service booking process
- Real-time notifications improve usability

Chapter 11

Future Scope

The Solwin system is designed as a scalable and flexible platform that can be enhanced with advanced features in the future. While the current system provides a secure and efficient solution for local services, there are several opportunities for further improvement and expansion.

11.0.1 1. Mobile Application Development

In the future, Solwin can be developed as a dedicated mobile application for Android and iOS platforms. This will improve accessibility and provide a better user experience through mobile-friendly features such as push notifications and real-time updates.

11.0.2 2. AI-Based Service Recommendation

Artificial Intelligence can be integrated to recommend the best service providers based on user preferences, past bookings, ratings, and location. This will improve decision-making and enhance user satisfaction.

11.0.3 3. Live Location Tracking

The system can be enhanced with real-time GPS tracking to allow users to monitor the exact location of service providers during travel. This will improve transparency and reduce waiting time.

11.0.4 4. Digital Payment Integration

Future versions can include secure online payment methods such as UPI, debit/credit cards, and digital wallets. This will enable cashless transactions and improve convenience for users.

11.0.5 5. QR Code Verification

A QR-based verification system can be introduced where customers can scan a provider's QR code at their doorstep to confirm identity instantly. This adds an extra layer of security.

11.0.6 6. Healthcare Integration (Care Module)

Solwin can be expanded to include a healthcare module that connects users with nearby hospitals and emergency services. In case of medical emergencies, user data such as blood group and medical history can be accessed quickly.

11.0.7 7. store module

collab with stores and give discout coupons for the customer who recharge the solwin payment patform for the payment

11.0.8 8. Advanced Admin Analytics

The admin panel can be enhanced with analytics and reports to monitor system usage, user behavior, and service trends. This helps in better decision-making and system improvement.

11.0.9 9. Expansion to Wider Areas

Currently focused on local areas, the system can be expanded to cover cities, districts, and eventually nationwide services.

11.0.10 10. Voice Assistant Integration

Future versions can include voice-based interaction, allowing users to book services using voice commands. This will improve accessibility for elderly users.

Chapter 12

Conclusion

The Solwin system was developed to address the major challenges present in the existing local service ecosystem, such as lack of security, transparency, and proper organization. The project successfully demonstrates how modern technology can be used to create a reliable and efficient platform for connecting customers with service providers.

12.0.1 Summary of Work

The system introduces a centralized digital platform that ensures safe and structured interaction between users and service providers. Key features such as **PCC verification**, **OTP-based service tracking**, and an **SOS emergency module** significantly improve the overall safety and reliability of the system.

The use of modern technologies like React.js and Supabase enabled the development of a responsive and scalable application. The modular design of the system ensures easy maintenance and future enhancements.

12.0.2 Achievement of Objectives

The main objectives of the project were successfully achieved:

- Providing a secure platform for local services
- Ensuring verification of service providers
- Improving transparency in service execution
- Enhancing user experience through a simple interface
- Introducing safety features like SOS alerts

12.0.3 Final Outcome

The final outcome is a fully functional working prototype that demonstrates all core features of the Solwin system. The platform effectively solves the limitations of the existing system and provides a better alternative for users.

Chapter 13

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